Networking HTB

Networking is a connection between two devices which shares information .

Networking enables two computers two communicate with each other.

wide area topologies - mesh/tree/star

medium- ethernet/fiber/coax/wireless

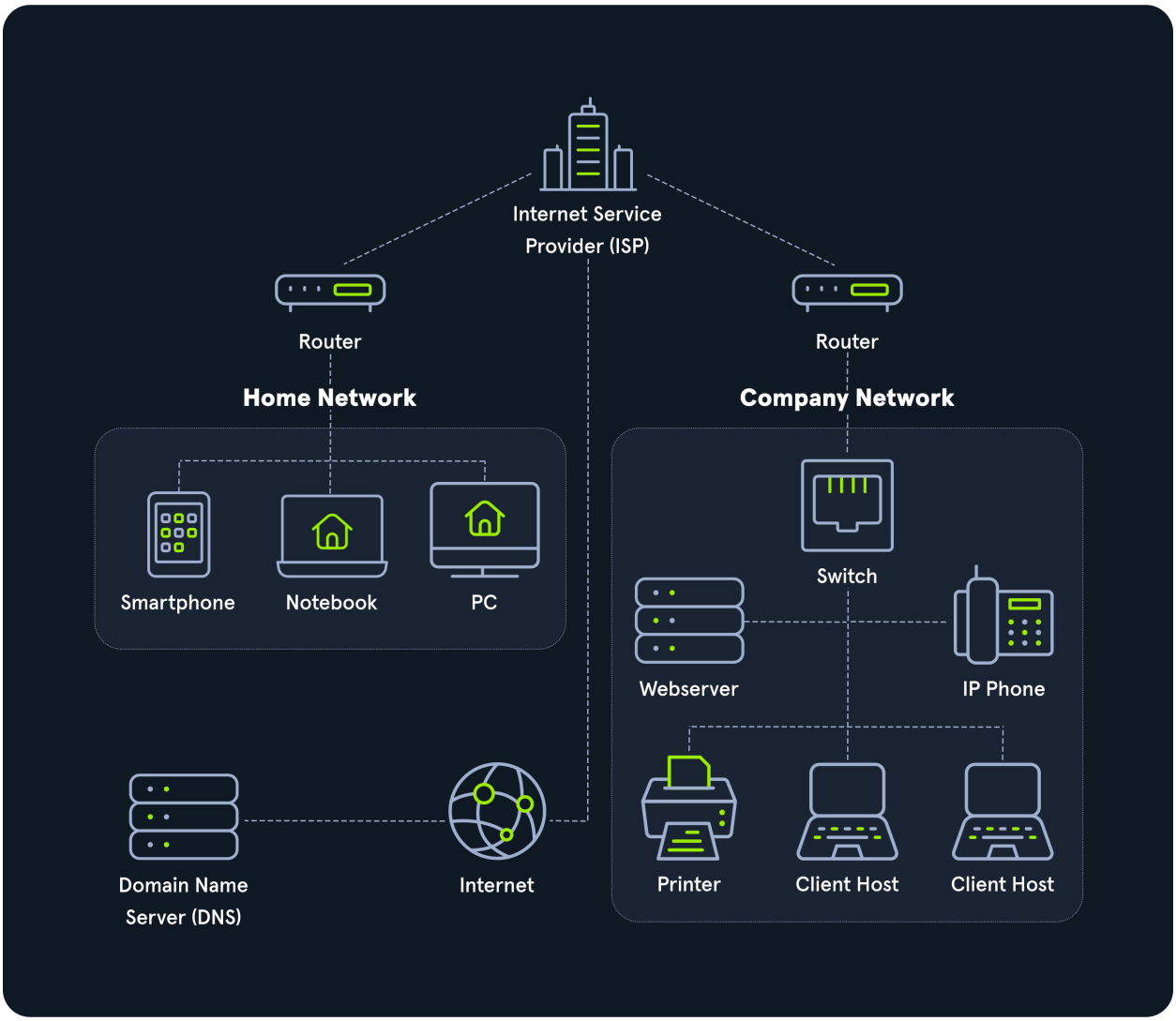
protocol-tcp/udp/ipx

*Adding Defence layer on network

By Creating lots of smaller Networks and having them communicate, we can add defence layers. by doing this it will make attacker slow down.

Most networks use a `/24` subnet, so much so that many Penetration Testers will set this subnet mask (255.255.255.0) without checking. The /24 network allows computers to talk to each other as long as the first three octets of an IP Address are the same (ex: 192.168.1.xxx). Setting the subnet mask to `/25` divides this range in half, and the computer will be able to talk to only the computers on "its half." We have seen Penetration Test reports where the assessor claimed a Domain Controller was offline when it was just on a different network in reality. The network structure was something like this:

- Server Gateway: 10.20.0.1/25
- Domain Controller: 10.20.0.10/25
- Client Gateway: 10.20.0.129/25
- Client Workstation: 10.20.0.200/25
- Pentester IP: 10.20.0.252/24 (Set Gateway to 10.20.0.1)



The diagram shows how networks are connected in a typical setup:

Main Components:

- **Internet Service Provider (ISP)**: At the top, this is the company that provides internet access to both networks.
- **Two Separate Networks**: The diagram shows a Home Network (left) and a Company Network (right), each with their own router connecting to the ISP.

Home Network includes:

- Smartphone, Notebook, and PC - all connected through the router

Company Network includes:

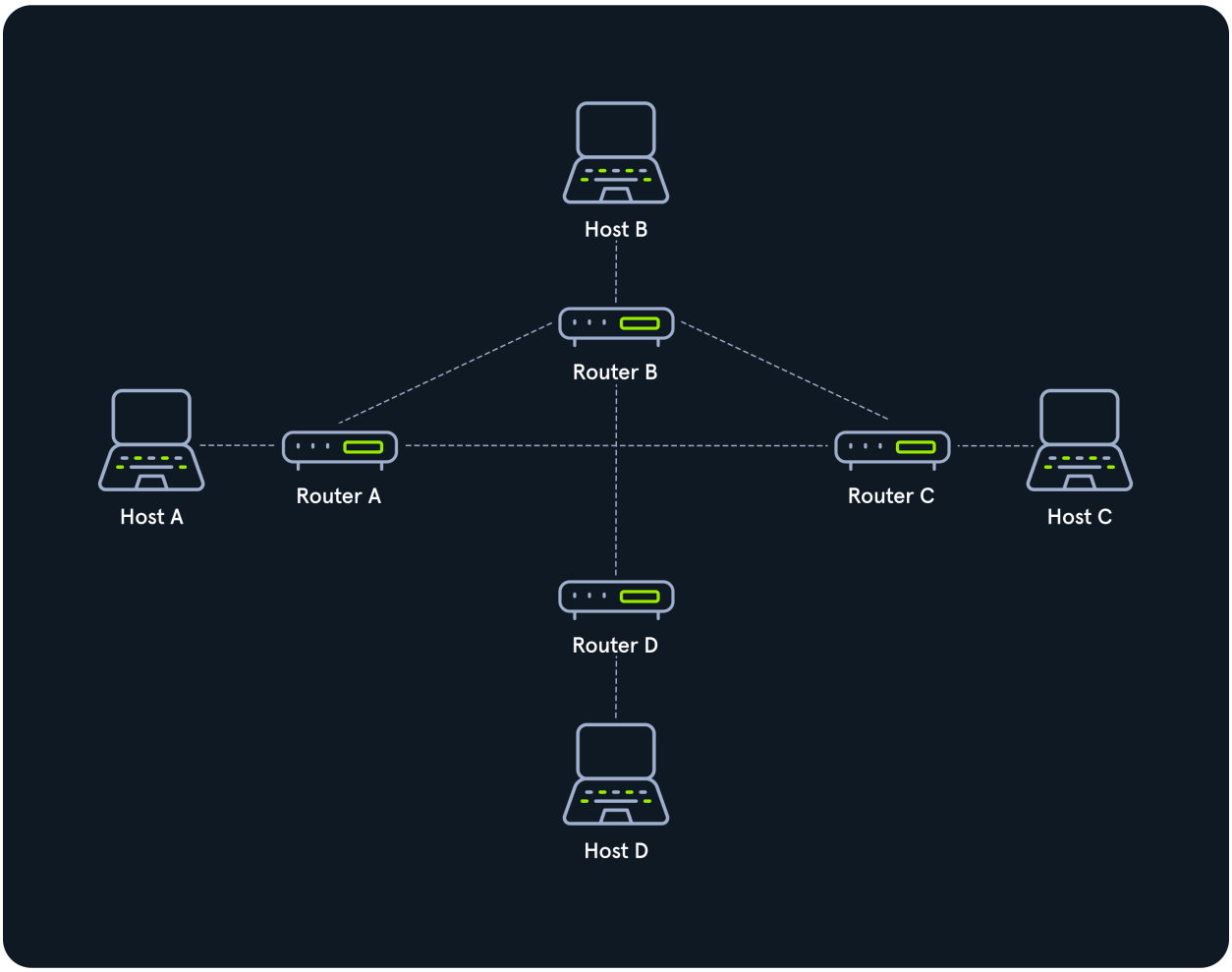
- A switch that connects multiple devices
- Webserver for hosting websites
- IP Phone for making calls over the internet
- Printer and multiple Client Host computers

Additional Elements:

- Domain Name Server (DNS): Translates website names into IP addresses
- Internet: Shown as a globe connecting everything together

This diagram illustrates how different networks are segmented for better security, which relates to the concept in your document about adding defense layers by creating smaller networks that communicate with each other.

Mesh topology



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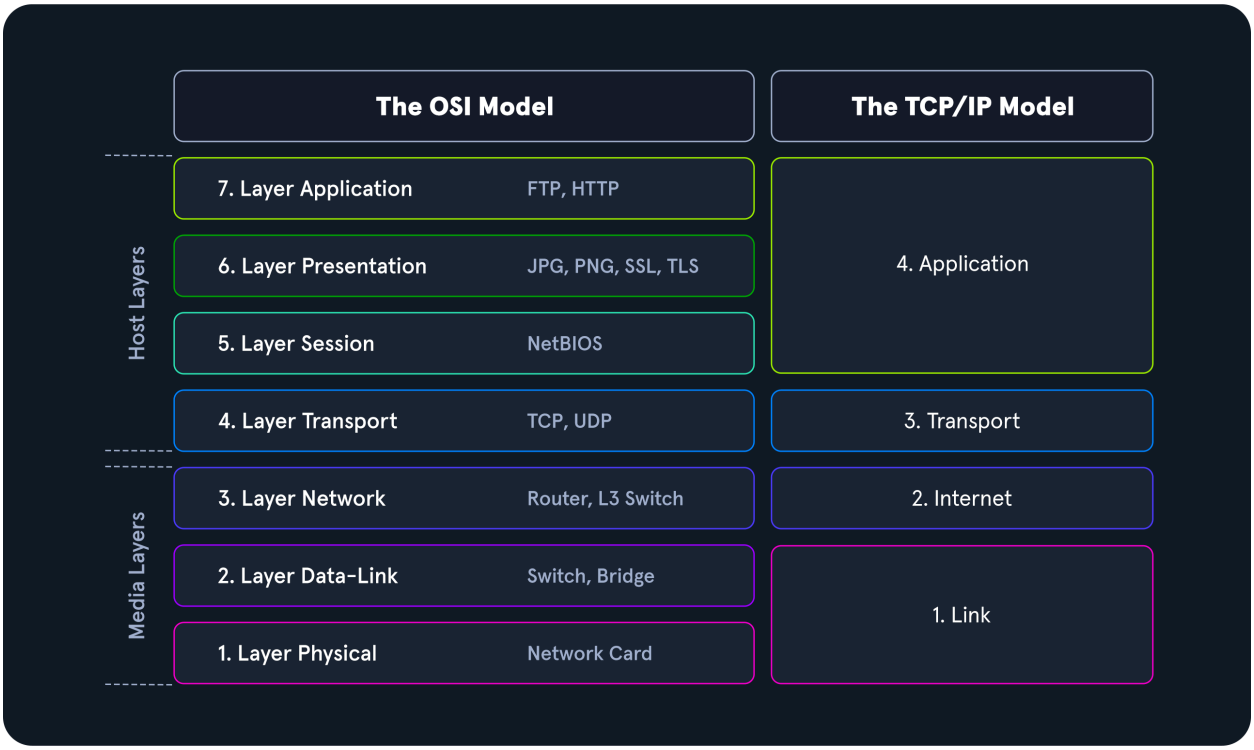
The image shows a mesh network topology, which is one of the wide area topologies mentioned in your document▸. In a mesh topology, each device (host) is connected to multiple other devices through routers, creating redundant paths for data transmission.

Proxy

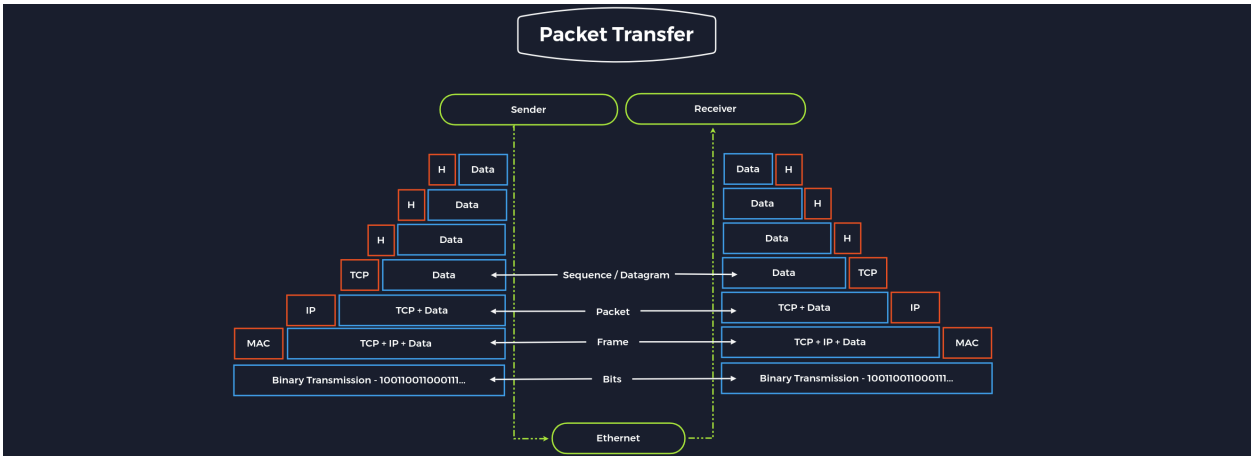
proxies will almost always operate at Layer 7 of the OSI Model

- Transparent Proxy
- Reverse Proxy
- Dedicated Proxy / Forward Proxy

OSI MODEL AND TCP/IP



- TCP/IP Model is responsible for switching and transport of data packets on the internet.
  tcp/ip is a communication protocol. that allow hosts to connect to the internet
-



The image shows how data is packaged and transferred between a sender and receiver in computer networking, illustrating the layered approach of the OSI and TCP/IP models.

Here's what the diagram explains:

- **Packet Transfer Process**: The diagram shows how data moves from a sender to a receiver through different layers of networking protocols.
- **Encapsulation**: As data moves down from the sender's application layer, each networking layer adds its own header (marked as "H", "TCP", "IP", "MAC") to the data. This process is called encapsulation.
- **Different Data Units**:
  - At the top layers: Raw data with headers
  - Sequence/Datagram: When TCP adds its header
  - Packet: When IP information is added
  - Frame: When MAC (physical address) information is added
  - Bits: The actual binary transmission that travels across the physical medium
- **Decapsulation**: When data reaches the receiver, the process is reversed - each layer strips off its corresponding header and passes the remaining data up to the next layer.
- **Physical Transmission**: At the bottom, everything is converted to binary (1s and 0s) for transmission across the physical medium (Ethernet in this case).

This diagram relates to the networking concepts in your document, particularly showing how the different layers of the TCP/IP model work together to transfer data across networks.

*To attack another system we have to need to know the IP of that target machine then only we have been able to taget that unless . we cant do anything by just knowing mac add its useless.

IP addresss make sure that the delivery of the data should be on correct address thats y we need ip.

There are two types of Ip - IPV4/IPV6.

`IPv4`, which consists of a `32`-bit binary number combined into `4 bytes` consisting of `8`-bit groups (`octets`) ranging from `0-255`. These are converted into more easily readable decimal numbers, separated by dots and represented as dotted-decimal notation.

CIDR

`Classless Inter-Domain Routing` (`CIDR`) is a method of representation and replaces the fixed assignment between IPv4 address and network classes (A, B, C, D, E). The division is based on the subnet mask or the so-called `CIDR suffix`, which allows the bitwise division of the IPv4 address space and thus into `subnets` of any size.
The `CIDR suffix` indicates how many bits from the beginning of the IPv4 address belong to the network. It is a notation that represents the `subnet mask` by specifying the number of `1`-bits in the subnet mask.

- IPv4 Address: `192.168.10.39`
- Subnet mask: `255.255.255.0`

🌐 What is Subnetting?

Subnetting = Splitting a big network into smaller networks (subnets).

Why?

- To **organize devices** efficiently.
- To **reduce network traffic** (less congestion).
- To **add security** (one subnet can't directly talk to another without a router/firewall).

📍 Example:

Imagine you own a company with **200 computers**.

If you put them all in **one network**:

- Every device can directly talk to every other device.
- Broadcast messages (like ARP) get sent to **all 200 devices** — that's noisy and slows down the network.

👉 Instead, you can **subnet** them:

- HR dept = Subnet A (50 devices)
- IT dept = Subnet B (80 devices)
- Sales dept = Subnet C (70 devices)

Now each subnet is **isolated**, and traffic is controlled by a router/firewall.

📍 How it Works Technically:

- IP addresses are **divided into 2 parts**:
  - **Network ID** → identifies the subnet
  - **Host ID** → identifies the device within that subnet

This division is controlled by the **Subnet Mask**.

📍 Subnet Mask

Example:

- IP: `192.168.1.25`
- Subnet Mask: `255.255.255.0`

Here:

- `255.255.255` = Network part (192.168.1)
- `.0` = Host part (last 8 bits → 256 possible addresses)

So:

- Network ID = `192.168.1.0`
- Host range = `192.168.1.1 – 192.168.1.254`

- Broadcast = `192.168.1.255`

📍 Why Hackers Care?

- Subnetting tells you **how many possible targets are in a network**.
- Example: If you compromise one computer, you'll check the subnet to know which other machines exist.
- Tools like `nmap` use this knowledge:

  ```
  nmap -sP 192.168.1.0/24
  ```

  This scans the **entire /24 subnet** (256 IPs).

⚡ In short:

Subnetting = **cutting a big IP range into smaller, logical groups** to control communication, reduce noise, and make recon more efficient.

Where subnetting gets easier (mental tricks):

1. **Know your "magic numbers"**

   In subnetting, the block size (how much each subnet "jumps") comes from this formula:

   ```
   Block size = 256 - (subnet mask value of the last octet)
   ```

   Example: `/28` → mask is `255.255.255.240`.

   Last octet = `240`.

   256 - 240 = **16**.

   So every subnet is a block of 16 IPs:

```
192.168.12.128
192.168.12.144
192.168.12.160
192.168.12.176
```

That way you don’t have to count in binary every time.